

Application of XR technology in stomatology education: Theoretical basis, application scenarios and future prospects

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ABSTRACT

Extended Reality (XR) technology is a new immersive technology based on Virtual Reality (VR), Augmented Reality (AR) and Mix Reality (MR). With the continuous development of science and technology, people are increasingly realizing that XR technology has practical significance in various fields, such as agriculture, medicine, education and other industries. We list some of the XR technologies currently in use in these areas and explain their role in promoting this area. At the same time, with people's attention on oral health, oral medicine is also getting more and more attention. We focus on the application of XR technology in stomatology education, analyze the theoretical basis of stomatology education and the application scenarios in the field of stomatology education, and put forward the prospect of the application of XR technology in the field of stomatology education.

Keywords: Virtual Reality, Augmented Reality, Mix Reality, Stomatology education.

Index Terms: Stomatology education—Augmented Reality—Virtual reality; Computing methodologies—Artificial intelligence—Knowledge representation and reasoning

1 INTRODUCTION

Extended reality (XR) technology includes technologies such as virtual reality (VR), augmented reality (AR) and mixed reality (MR) [1]. XR technology is a new technology based on VR, AR and MR, which has the characteristics of situational awareness, sensory substitution, intuitive interaction and editing reality [2]. In recent years, with the continuous development of information technology and greatly improved computing power, XR technology has been developed and applied in many fields such as agriculture, medicine, education and military [3].

One area where XR has been used productively in sports is to help develop tactical skills or train athletes in tactical decision making. XR tools can be developed to improve the tactical education courses currently used for education and training athletes in tactical decision making, which can be carried out using VR technology [4]. At the same time, XR tools can be used to carry out some perceptual tasks, such as simulating the real world running of rugby players, and importing the real-world kinematic information captured in the real-world environment into the VR system to enhance the simulation. XR tools can also create perceptual information, trying to recreate information from the real world, like creating soccer tracks. On the basis of XR virtual simulation technology, the hospital has completed the production of virtual

mouth by referring to the morphological structure and characteristics of Chinese teeth [5]. XR technology can also simulate tooth preparation and determine the amount of preparation, and cut on different planes, so that medical students can feel the closeness between different incision depths and pulp, so as to consolidate and strengthen the theoretical knowledge of oral teaching, and guide experimental teaching and clinical practice with more intuitive visual feelings.

In medicine, the impact of extended reality is widespread. Virtual reality platforms are designed to teach basic subjects like human anatomy and even train surgeons to perform complex surgical procedures [6]. Augmented reality and mixed reality offer new ways to improve surgical safety and success by visualizing intraoperative procedures and diagnostic images with devices such as Google Glass [7]. In medicine, virtual reality's ability to move patients away from their physical environment also offers treatments for rehabilitation, pain, and mental disorders. In spinal medicine, MR enables the surgeon to obtain anatomical information about the patient intraoperatively and superimpose virtual holographic elements on the surface anatomy of the actual patient in real time on the operating table, thus enabling holographic navigation. XR technology can reproduce human body structure, pathophysiology, and clinical scenes with a sense of reality, and is of great significance in medical education, surgical simulation and planning, intraoperative guidance and other scenarios [8].

With the continuous development of science and technology and the continuous improvement of people's living standards, XR technology has gradually covered various fields in our real lives and played a significant role. Nowadays, people have higher requirements for the quality of life, which also promotes the development of XR technology in oral education and other industries. Next, we will introduce XR technology, theoretical background of stomatology education and application of XR technology in stomatology education in the following sections.

2 EXTENDED REALITY TECHNOLOGY

Extended reality (XR) refers to a human-computer interactive environment generated by combining physical environment with virtual environment through computer technology and wearable devices. Extended reality technology can be regarded as an umbrella term, including virtual reality technology (VR), augmented reality technology (AR), mixed reality technology (MR) and other new immersive technologies that may appear due to technological progress [9]. Virtual reality technology is a multi - information fusion, interactive, three - dimensional dynamic real - scene computer simulation system. Virtual reality technology, dominated by sound and vision, uses computers to simulate virtual environments and immerse users in the environment [10]. In this

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simulation system, users must wear VR glasses or VR headset to obtain 360-degree view of the simulated world, as shown in Figure 1. Augmented reality technology refers to a new technology that integrates real world information and virtual world information "seamlessly". It is to superimpose physical information (visual information, sound, taste, touch, etc.) within a certain time and space of the real world through computer science and technology simulation, and then map the obtained virtual information to the real world, which can be perceived by human senses. To achieve a sensory experience that transcends reality. Real environments and virtual objects are superimposed in real time onto the same frame or space. Mixed reality technology is a further development of VR and AR, which digitizes the physical world and integrates it with the virtual world to create a new world, in which virtual objects and objects in the real world coexist and can interact in real time [11].



Figure 1. Left to right: (a) Augmented Reality (AR) and (b) Mixed Reality (MR) experiences extend reality by adding virtual elements, whereas (c) a Virtual Environment (VE) can be a fantasy world, and (d) a Virtual Reality (VR) immerses the user in a "real" setting such as a travel experience.

The characteristics of XR are as follows. (1) Near-eye display. Near-eye display technology is the premise of improving immersion. Near-eye display is the realization platform of virtual reality and augmented reality. Near-eye displays promise to revolutionize healthcare, communications, entertainment, education, manufacturing and more. The ideal near-eye display must be able to provide high-resolution images in a large field of view while supporting adaptive prompts and large eye masks, and have a compact contour size; (2) Real-time content production. From the perspective of interaction between users and content applications, virtual reality services can be divided into weak interaction and strong interaction. The former is mainly used for passive viewing of panoramic video on demand and live broadcast, while the latter is commonly used for games, interactive education and other forms. The content must be rendered in real time according to the interactive information input by users, with stronger freedom, real-time performance and sense of interaction; (3) perceptual interaction. Perceptual interaction emphasizes technological collaboration with other fields. At present, many perceptual interaction technologies, such as tracking and positioning, immersive sound field, gesture tracking, eye tracking, three-dimensional reconstruction, machine vision, electromyography sensing, speech recognition, smell simulation, virtual movement, tactile feedback, and brain-computer interface, are flourishing, co-existing and complementary, and have their own advantages in each subdivided scene. In the future, naturalization, contextualization and intellectualization will become the main direction of perceptual interaction technology. (4) fixation point rendering. Rendering processing mainly involves two parts. One is content rendering, which is the process of projecting 3D virtual space scenes onto the plane to form plane images in the process of content production. The second is terminal rendering, that is, the processing process of optical distortion and dispersion correction for the plane image generated by content rendering and frame insertion according to the user's attitude. All of the rendering techniques are designed to improve rendering performance and achieve user-perceived details at higher resolution with a small overhead. Among them, the key of VR rendering lies in complex content calculation. The rendering technology of AR and MR is

basically the same as that of VR, but the application scene focuses on the integration with the real world, such as virtual-real occlusion, light and shadow rendering, material reflection rendering, etc. [12].

3 THEORETICAL BASIS OF STOMATOLOGY EDUCATION BASED ON XR TECHNOLOGY

With the rapid development of society, the self-protection consciousness of hospital patients is becoming stronger and stronger, the occurrence frequency of doctor-patient problems is becoming higher and higher, and the contradiction between doctors and patients is becoming increasingly severe. Therefore, medical students must master various operational skills before formal clinical practice, which puts forward higher requirements for the simulation of clinical environment. Under this background, the integration of XR technology and medical education comes into being.

The laboratory of the University of California, Los Angeles has long developed the project of "human brain engineering", the European strategic plan of information technology development and development includes a plan of "comprehensive human animation model library" related to medicine, and the Medical School of Hamburg University has also constructed a set of human virtual reality system [13]. Compared with foreign countries, China's XR technology started relatively late, but a number of domestic medical universities have successfully carried out a number of clinical operations based on this technology. In addition to the research on the whole human organs, some large hospitals and scientific research institutes in China have also actively carried out the XR technology research on oral organs. For example, XR technology is used to conduct 3D image simulation for the design scheme of oral implant repair, and computer software is used to analyze and evaluate the biological rationality of the design scheme, so as to realize the teaching function of virtual image simulation and therapeutic effect evaluation of oral therapy scheme and implant surgery process. On the basis of traditional teaching, virtual teaching of XR technology can realize teaching functions that traditional experiments do not have and are difficult to complete, such as irreversible operation and high-cost comprehensive training [14].

In the process of practice, many hospitals also use XR technology for reform, so as to overcome the problem of empty language in traditional oral teaching, as well as the problem of abstract demonstration of traditional dental arrangement techniques. XR technology transforms the abstract and fuzzy knowledge that students must master into a three-dimensional simulation model by means of specific operations, which is conducive to the memory and learning of oral knowledge for medical students and enables them to have a more intuitive experience of the making and arrangement of teeth. At the same time, XR virtual teaching also aims at the current domestic oral theory is not easy to understand, it is difficult to collect real teeth in experimental teaching, operation details cannot take into account every student and other practical problems, creating a clearer three-dimensional teaching environment for medical students.

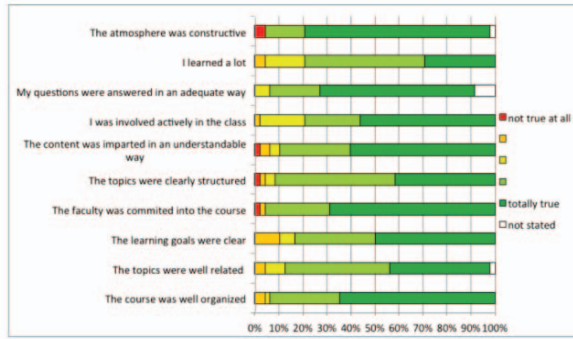


Figure 2. Students' responses to first ten question on questionnaire about the hands-on module

At present, there has been a credible theoretical support for XR technology oral education. For example, some foreign scholars compared the knowledge grasp of third-grade dental students after the conventional teaching of zirconia full crown making and the group's hands-on CAD/CAM digital module teaching [15], as shown in Figure 2. The results show that the CAD/CAM group is significantly higher than the conventional group. It can be seen that using CAD/CAM digital single-crown model to design zirconia full crown is more conducive to students' mastery of knowledge. XR digital teaching of stomatology can make stomatology professional skills training simpler and more standard. Digital teaching is synchronized with traditional classroom teaching in real time. Open after-school exercises can be set in different stages according to the key and difficult points, which makes it easier for students to get started and feedback the teaching effect in time.

In addition to the above theoretical support, the interaction of XR technology is also evolving. XR oral digital teaching has always insisted on building a large learning circle, and developing small learning circles with different characteristics of medical students to ensure their sense of belonging and interaction. For example, interns mainly use the form of case discussion, undergraduates use the form of multimedia courseware and open homework, graduate students use the form of thematic discussion. At the same time, digital oral teaching can also cross with other teaching contents, which is not only conducive to the establishment of interactive learning environment matching with students' thinking, but also conducive to the improvement of oral content and the improvement of oral teaching classroom quality.

4 APPLICATION SCENARIOS OF XR TECHNOLOGY IN THE FIELD OF STOMATOLOGY EDUCATION

In the field of stomatology education, the traditional teaching method of stomatology is mainly classroom teaching. Teachers conduct classroom teaching through blackboard writing, oral teaching and other methods, combined with auxiliary tools such as human oral model, oral specimen, human skeletal model and wall chart. The traditional indoctrination teaching method is not easy to understand, and low efficiency. In addition, due to the complexity of oral environment and human function, the basic theories and knowledge related to oral medicine that medical students need to master, as well as the whole process of surgical clinical practice and surgical training, will be subject to various restrictions and obstacles due to factors such as time, space and technology.

With the rapid development of Internet and communication technology, especially in the medical field in recent years, many cutting-edge communication and computer technologies have been fully practiced and applied. For example, China Mobile Jiangsu Branch, with its mature and advanced 5G technology and high coverage rate, realized a series of intelligent medical applications such as remote consultation, hospital emergency and medical cloud

platform, which to a large extent alleviated the shortage of medical resources and the difficulty of seeing a doctor.

Emerging cutting-edge science and technology have enabled traditional medical industry, transformed the Internet from an application platform into a man-made resource, integrated it into an important part of medical diagnosis and rescue procedures, solved long-standing pain points, and realized a real medical revolution. On this basis, more and more scholars and experts have shifted their attention and energy to the field of integrating Extended Reality technology with traditional medicine. Andrews and Southworth [16] fully affirmed extended reality technology and proved that ongoing work to apply extended reality technology to medical education and training on the ground is promising. Kim et al. [17] believe that extended reality technology can provide immersive and real user experience, which is feasible for training and real-time guidance in unconventional tasks, diagnosis, intervention and program application. Mastro et al. [18] believe that extended reality technology can provide additional and effective guidance for novices, in general, to improve their ability to make medical decisions, use medical equipment and perform medical procedures.

In the past, our country focused on the basic theoretical knowledge of oral medicine, which cannot provide the students with the conditions of observing the mechanism and processes of the incidence of oral diseases in real time. The teaching content is abstract and far from field practice, which is some difficulties for the students to grasp the related knowledge of oral medicine. However, in the process of traditional oral medicine practice teaching, the imitation head model system is mainly used for guidance and training, but this teaching system has obvious defects, unable to feedback patients' nerve pain and various discomfort caused by treatment in the process of treatment, resulting in students unable to improve their wrong operation behavior in the process of treatment due to excessive discomfort of patients. In addition, the scope of systematic training and teaching of the head model is limited, which cannot well simulate other oral related diseases, such as oral and maxillofacial soft tissue injury, tooth pulp disease, etc. The nerves and blood vessels in the oral cavity of human body are intricate. In order to reduce the incidence of surgical errors and accidents, it is necessary to provide students with a dynamic model for real-time observation and feedback from an all-round perspective in the teaching process. Extended reality technology fits the teaching needs of stomatology completely.

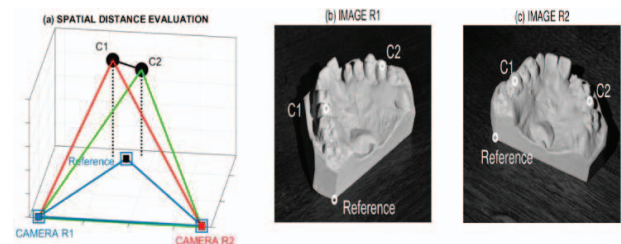


Figure 3: The determination of the distance between specified objects in space

The application of extended reality to traditional oral medical teaching and training has the following benefits. First, extended reality technology can conduct real-time 3D modeling of human oral environment, so that students can observe the change mechanism of oral environment and the root causes of lesions from a comprehensive perspective in the process of learning and training. For example, it overcomes the defect that only two-dimensional perspective can be displayed in wall chart teaching, avoids the understanding deviation caused by insufficient spatial imagination ability in the learning and training process of students, and thus significantly reduces students' mistakes in the practical operation process. Secondly, extended reality technology can break the

restrictions of time, space and region, has the characteristics of real-time operation and high safety, reduces the consumption of medical resources, alleviates the shortage of teaching resources to a large extent, and avoids ethical conflicts to a certain extent. Finally, in the process of teaching and training, it can shorten the time of clinical training for students, save training costs, and minimize hospital consumables and patient pain.

With the extensive application and development of extended reality technology in the field of stomatology teaching, the diagnosis and treatment operation training system with more complete functions and strong operability combined with this technology has been constantly introduced. The simulated oral simulation examination, oral profile examination and dental preparation system developed by Peking University Stomatological Hospital have combined virtual simulation technology with traditional teaching methods. It overcomes the limitation of traditional teaching method and realizes the innovation breakthrough of oral medicine experimental teaching method driven by independent research. Aleš Procházka et al. [19] discussed the use of augmented reality digital methods in stomatology, such as intraoral scanning for the acquisition and evaluation of dental arch data. As in Figure 3, the approach includes spatial object analysis using digital filters and morphological manipulation, as well as assessment of changes during treatment for specific diseases. Although there are still technical obstacles and various uncertainties, it can be seen from the research and conclusion of many scholars that the way of combining extended reality technology with traditional medicine teaching and stomatology teaching is quite optimistic. With the continuous iteration and maturity of extended reality technology, Extended reality technology will also be more widely and deeply applied in the field of medical teaching, and students can fully feel the immersive virtual practical experience in this "immersive" teaching process built by extended reality technology.

5 PROSPECT OF FUTURE STOMATOLOGY EDUCATION MODEL BASED ON XR TECHNOLOGY

At present, the XR technology is still in the starting stage in the field of oral medicine education. With the continuous development of national economy, the demand for oral medical education in the future will also improve. Virtual reality, augmented reality, and mixed reality use a variety of different software and hardware, but they share three main characteristics: immersion, presence, and interactivity. XR is a combination of interactive and immersive technologies that can add to or change our real-world experiences. Therefore, the application of XR technology has a strong practical significance for improving the mode of oral medicine education.

XR technology is human, intelligent, interactive, generative and immersive, which can help simplify learning, training and research processes. With XR technology, people can feel the virtual representation and experience it in an immersive 3D environment. This can effectively promote the integration of situation and knowledge, and can significantly improve the learning memory and teamwork of students and trainees. There are bound to be more XR educational products in the future that use virtual reality to enhance the breadth and depth of the learning experience, allow teachers and students to put themselves in each other's shoes, and also help teachers and students to foster a higher level of communication, which in turn reduces bias in the imparting of knowledge. The emergence of these products may change the future model of oral medicine education and make it more intelligent.

At the same time, in the field of stomatology education, which involves a large number of medical training and experiments, the emergence of XR technology can help simulate many realistic scenes, reconstruct the environment difficult to simulate in training,

provide a safer training environment, and reduce the risk of injury and damage of expensive equipment. XR's benefits also include the ability to control and manipulate constraints in complex and dynamic environments to create repeatable specific situations [20]. Therefore, XR technology can save a lot of real resource consumption and help to provide a more convenient way of training. At present, XR technology may not be widely applied in the field of stomatology education. However, with the large-scale application and technological update and iteration, it is expected that XR technology will become more and more popular, and XR technology will be a part of the future field of stomatology education.

In the future, the society will have more demands for stomatology education. The application of XR technology may help students or scholars in related fields to conduct research more conveniently and improve their efficiency. At the same time, XR technology can also be used to popularize relevant medical knowledge, so that more people can improve the understanding of oral health, and help to solve the serious imbalance between doctors and patients. In the field of oral medicine education, XR technology is a new technology, although there are still limitations such as high cost and low popularity. With the continuous development of technology, large-scale use, analysis of the development of XR technology in other fields, the introduction and development of related XR technology in the field of stomatology education is a big trend. Perhaps, if XR technology intersects with it, it will create more niche areas worth investigating.

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